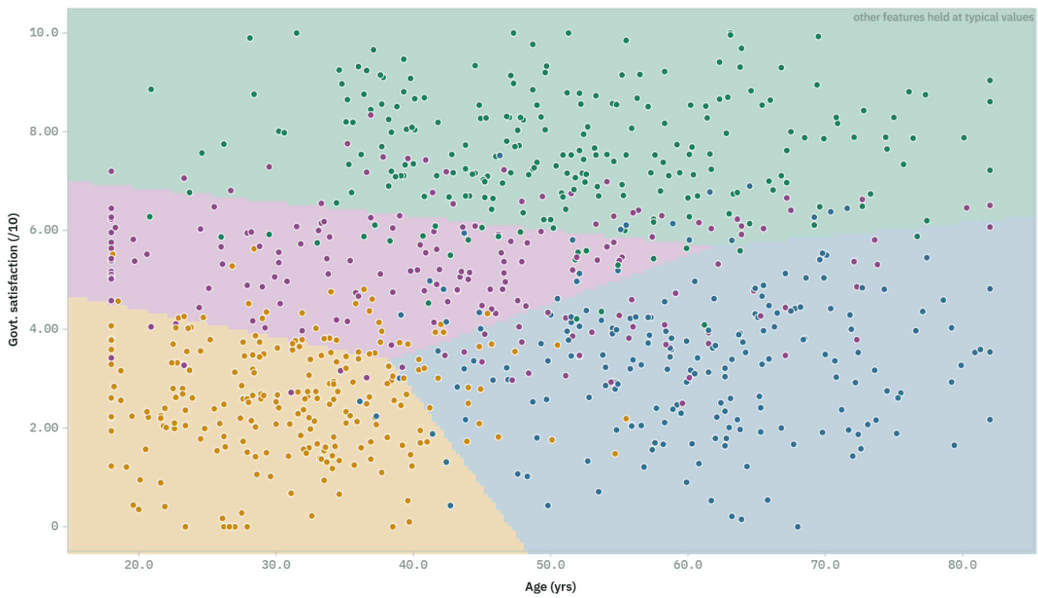
Testleaf AI Playground for ML

Machine Learning · Classification

ALGORITHM | Linear Regression | Classification | DATASET | ‹ | Election Preference (Multiclass) | ›

trained · 672/168 split | Train again | Upload data



● Party A ● Party T ● Party C ● Undecided

background = the winner in that area

ROWS TRAINED ON

672

80% of the data

ROWS TESTED ON

168

held back, never seen

WHAT THIS DATA IS

Survey data with four possible answers, not two. Age and satisfaction with the current government drive the choice, with city voters leaning differently from rural ones. Parties are fictional — A, T, C — exactly as in the slide.

MAKE A PREDICTION

Age 29.0 yrs

Govt. satisfaction 2.50 /10

Urban voter | Yes | No | Yes

Predict

PREDICTED PREFERENCE **Party T**

Party A

0.0%

Party T

97.7%

Party C

0.6%

Undecided

1.7%

```
Step 1 — each class gets its own score = base +
(weight × this voter's value), added up

  Party A
    -10.1 pts ← base points
    + 0.025 × 29.0 (Age) = +0.721
    + 1.84 × 2.50 (Govt. satisfaction) = +4.60
    − 1.66 × Yes (Urban voter) = -1.66
    ———
    = -6.45 pts

  Party T
    12.6 pts ← base points
    − 0.126 × 29.0 (Age) = -3.65
    − 1.70 × 2.50 (Govt. satisfaction) = -4.25
    + 0.496 × Yes (Urban voter) = +0.496
    ———
    = 5.23 pts

  Party C
    1.93 pts ← base points
    + 0.101 × 29.0 (Age) = +2.94
    − 1.11 × 2.50 (Govt. satisfaction) = -2.78
    − 1.94 × Yes (Urban voter) = -1.94
    ———
    = 0.155 pts

  Undecided
    2.45 pts ← base points
    − 0.023 × 29.0 (Age) = -0.666
    + 0.150 × 2.50 (Govt. satisfaction) = +0.376
    − 0.987 × Yes (Urban voter) = -0.987
    ———
    = 1.17 pts
———
highest score wins → Party T
```

```
Step 2a — convert each score to e^score (always
positive)

  Party A     e^-6.45 = 0.002
  Party T     e^5.23 = 186
  Party C     e^0.155 = 1.17
  Undecided   e^1.17 = 3.22
———
Total (add all 4 up)  =  190
```

```
Step 2b — each class's share = its own e^score ÷
that same total (190)

  Party A     0.002 ÷ 190 = 0.0%
  Party T     186 ÷ 190 = 97.7%
  Party C     1.17 ÷ 190 = 0.6%
  Undecided   3.22 ÷ 190 = 1.7%
———
All 4 shares add back up to 100% — that's the
whole point of dividing by the same total. e (≈
2.71828) is just a fixed constant that turns any
```

score positive and stretches the gap between a high score and a low one.

THE TRAINED MODEL

Party A score = **-10.1** pts ← base points (starting score) + **0.025**×Age + **1.84**×Govt. satisfaction − **1.66**×Urban voter

Party T score = **12.6** pts ← base points (starting score) − **0.126**×Age − **1.70**×Govt. satisfaction + **0.496**×Urban voter

Party C score = **1.93** pts ← base points (starting score) + **0.101**×Age − **1.11**×Govt. satisfaction − **1.94**×Urban voter

Undecided score = **2.45** pts ← base points (starting score) − **0.023**×Age + **0.150**×Govt. satisfaction − **0.987**×Urban voter

SAMPLE CALCULATION — ONE REAL ROW

45.7 yrs · 6.43 /10 · not urban voter

Party A	**2.86**
Party T	**-4.05**
Party C	**-0.580**
Undecided	**2.37**

highest wins → Party A
actual Party A

With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.

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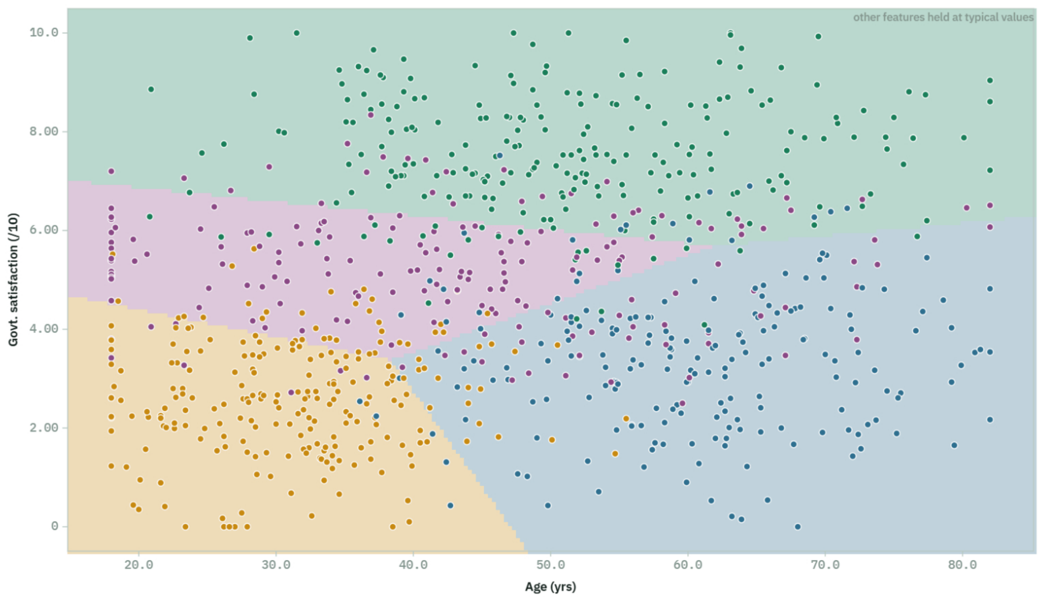

● Party A ● Party T ● Party C ● Undecided

background = the winner in that area

ROWS TRAINED ON

672

80% of the data

ROWS TESTED ON

168

held back, never seen

WHAT THIS DATA IS

Survey data with four possible answers, not two. Age and satisfaction with the current government drive the choice, with city voters leaning differently from rural ones. Parties are fictional — A, T, C — exactly as in the slide.

MAKE A PREDICTION

Age — 66.0 yrs

Govt. satisfaction — 2.50 /10

Urban voter — Yes | No — No

Predict

PREDICTED PREFERENCE **Party C**

Party A

0.0%

Party T

0.3%

Party C

98.6%

Undecided

1.1%

```
Step 1 — each class gets its own score = base +
(weight × this voter's value), added up

  Party A
    -10.1 pts ← base points
    + 0.025 × 66.0 (Age) = +1.64
    + 1.84 × 2.50 (Govt. satisfaction) = +4.60
    − 1.66 × No (Urban voter) = +0

    = -3.87 pts

  Party T
    12.6 pts ← base points
    − 0.126 × 66.0 (Age) = -8.31
    − 1.70 × 2.50 (Govt. satisfaction) = -4.25
    + 0.496 × No (Urban voter) = +0

    = 0.070 pts

  Party C
    1.93 pts ← base points
    + 0.101 × 66.0 (Age) = +6.68
    − 1.11 × 2.50 (Govt. satisfaction) = -2.78
    − 1.94 × No (Urban voter) = +0
```


1.1%

```
Step 1 — each class gets its own score = base +
(weight × this voter's value), added up

  Party A
    -10.1 pts ← base points
    + 0.025 × 66.0 (Age) = +1.64
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    + 0.496 × No (Urban voter) = +0

    = 0.070 pts

  Party C
    1.93 pts ← base points
    + 0.101 × 66.0 (Age) = +6.68
    − 1.11 × 2.50 (Govt. satisfaction) = -2.78
    − 1.94 × No (Urban voter) = +0
```


```
  Party A     0.021 ÷ 348 = 0.0%
  Party T     1.07 ÷ 348 = 0.3%
  Party C     343 ÷ 348 = 98.6%
  Undecided   3.70 ÷ 348 = 1.1%

All 4 shares add back up to 100% — that's the
whole point of dividing by the same total. e (≈
2.71828) is just a fixed constant that turns any
```

2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

THE TRAINED MODEL

Party A score = **-10.1** pts ← base points (starting score) + **0.025**×Age + **1.84**×Govt. satisfaction − **1.66**×Urban voter

Party T score = **12.6** pts ← base points (starting score) − **0.126**×Age − **1.70**×Govt. satisfaction + **0.496**×Urban voter

Party C score = **1.93** pts ← base points (starting score) + **0.101**×Age − **1.11**×Govt. satisfaction − **1.94**×Urban voter

Undecided score = **2.45** pts ← base points (starting score) − **0.023**×Age + **0.150**×Govt. satisfaction − **0.987**×Urban voter

SAMPLE CALCULATION — ONE REAL ROW

45.7 yrs · 6.43 /10 · not urban voter

Party A	**2.86**
Party T	**-4.05**
Party C	**-0.580**
Undecided	**2.37**

highest wins → Party A
actual Party A

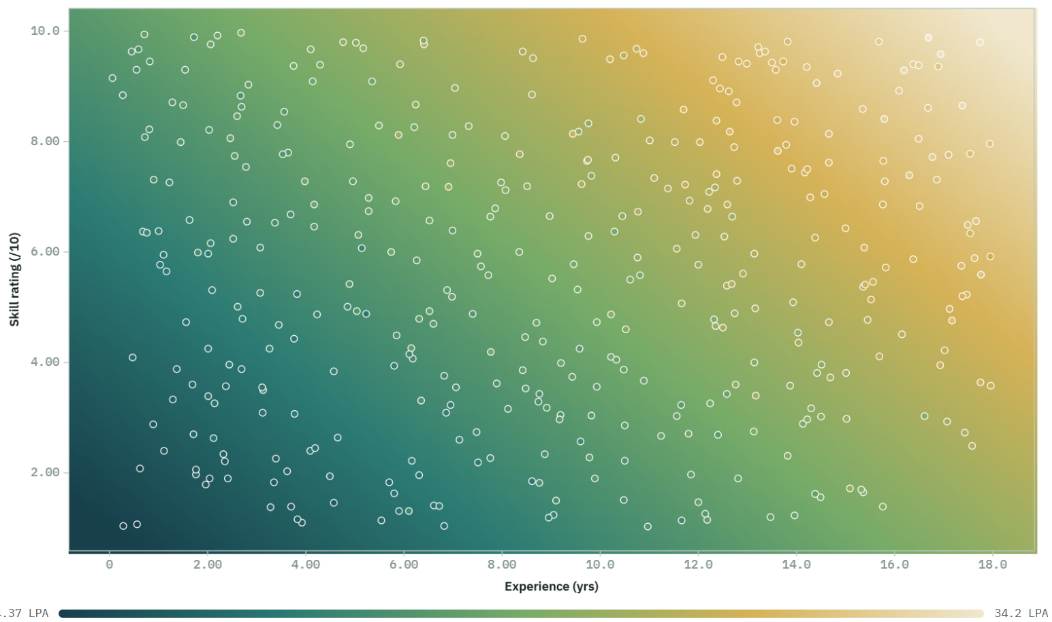
With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.

Testleaf AI Playground for ML

Machine Learning · Linear Regression

ALGORITHM **Linear Regression** Classification DATASET ‹ QA Engineer Salary ›

trained · 320/80 split **Train again** Upload data



ROWS TRAINED ON

320

80% of the data

ROWS TESTED ON

80

held back, never seen

WHAT THIS DATA IS

Experience and skill rating each add a predictable amount to the package.

MAKE A PREDICTION

Experience 6.00 yrs

Skill rating 7.00 /10

Predict

TOTAL PACKAGE **18.3 LPA**

```
Base salary — the starting value, before any
feature counts
  3.10 LPA

Experience
  0.966 LPA per year × 6.00 yrs
  = +5.80 LPA

Skill rating
  1.34 LPA per point × 7.00 /10
  = +9.39 LPA

TOTAL   18.3 LPA
```

THE TRAINED MODEL

```
Salary_LPA in LPA =
  3.10 LPA ← base salary (minimum)
  + 0.966 LPA per year  × Experience
  + 1.34 LPA per point  × Skill rating
```

SAMPLE CALCULATION — ONE REAL ROW FROM THE DATA

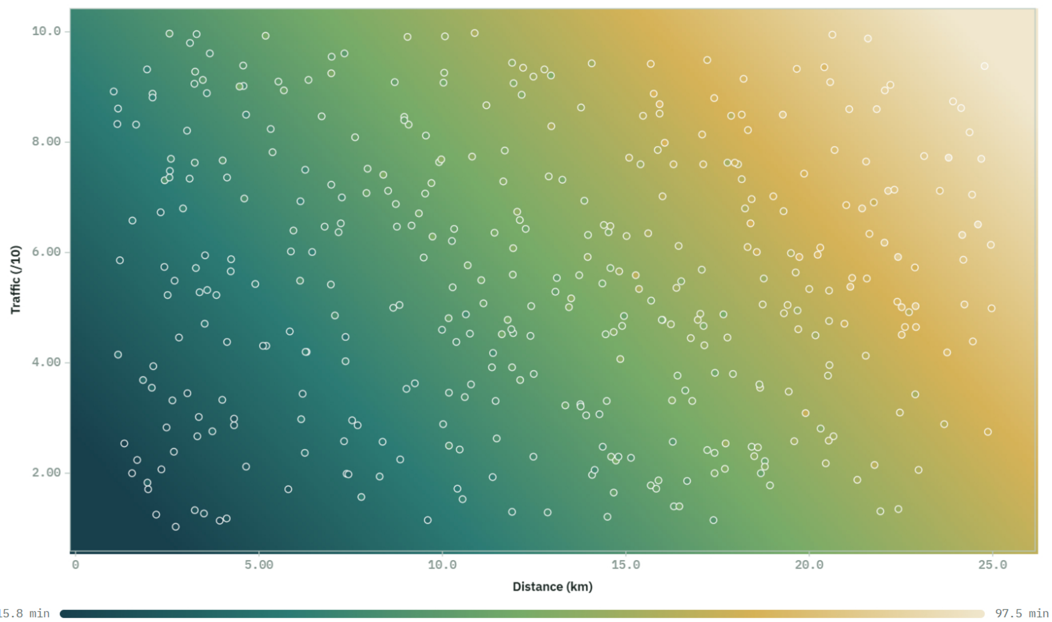
```
9.46 yrs · 5.78 /10

  3.10 (base)
  + 0.966 × 9.46 = +9.14
  + 1.34 × 5.78 = +7.75

model says   20.0 LPA
actual value  19.4 LPA
```

Read it as a rate card. The first number is the **base salary** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.

Testleaf AI Playground for ML Machine Learning · Linear Regression

ALGORITHM Linear Regression Classification DATASET ‹ Food Delivery Time ›

trained · 320/80 split Train again Upload data



ROWS TRAINED ON
320
80% of the data

ROWS TESTED ON
80
held back, never seen

WHAT THIS DATA IS

Distance and traffic both push the delivery time up at a steady rate.

MAKE A PREDICTION

Distance 8.00 km

Traffic 6.00 /10

Predict

TOTAL TIME 45.7 min

```
Base time — the starting value, before any
feature counts
  7.56 min

Distance
  2.40 min per km × 8.00 km
  = +19.2 min

Traffic
  3.16 min per point × 6.00 /10
  = +18.9 min

TOTAL   45.7 min
```

THE TRAINED MODEL

```
Delivery_Minutes in min =
  7.56 min ← base time (minimum)
  + 2.40 min per km  × Distance
  + 3.16 min per point  × Traffic
```

SAMPLE CALCULATION — ONE REAL ROW FROM THE DATA

```
8.96 km · 8.46 /10

  7.56 (base)
  + 2.40 × 8.96 = +21.5
  + 3.16 × 8.46 = +26.7

model says   55.8 min
actual value   55.4 min
```

Read it as a rate card. The first number is the **base time** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.